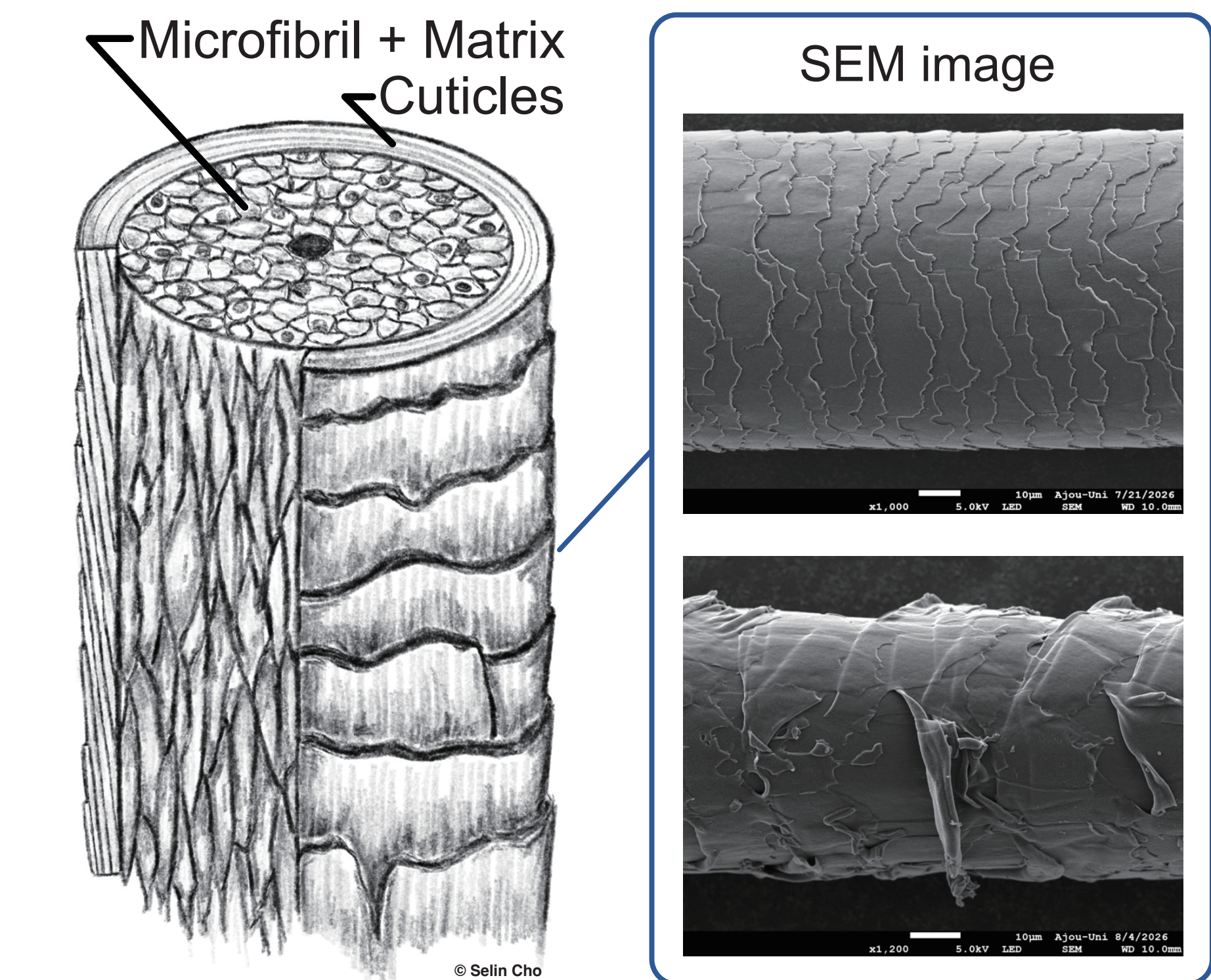


What is hair?

- Human hair is a hygroscopic, hierarchically structured filament composed of cortex surrounded by a cuticle layer.
- The cortex contains microfibrils embedded in a hygroscopic matrix, and the asymmetric distribution of these components give rise to intrinsic curvature characteristics.



Hair frizz

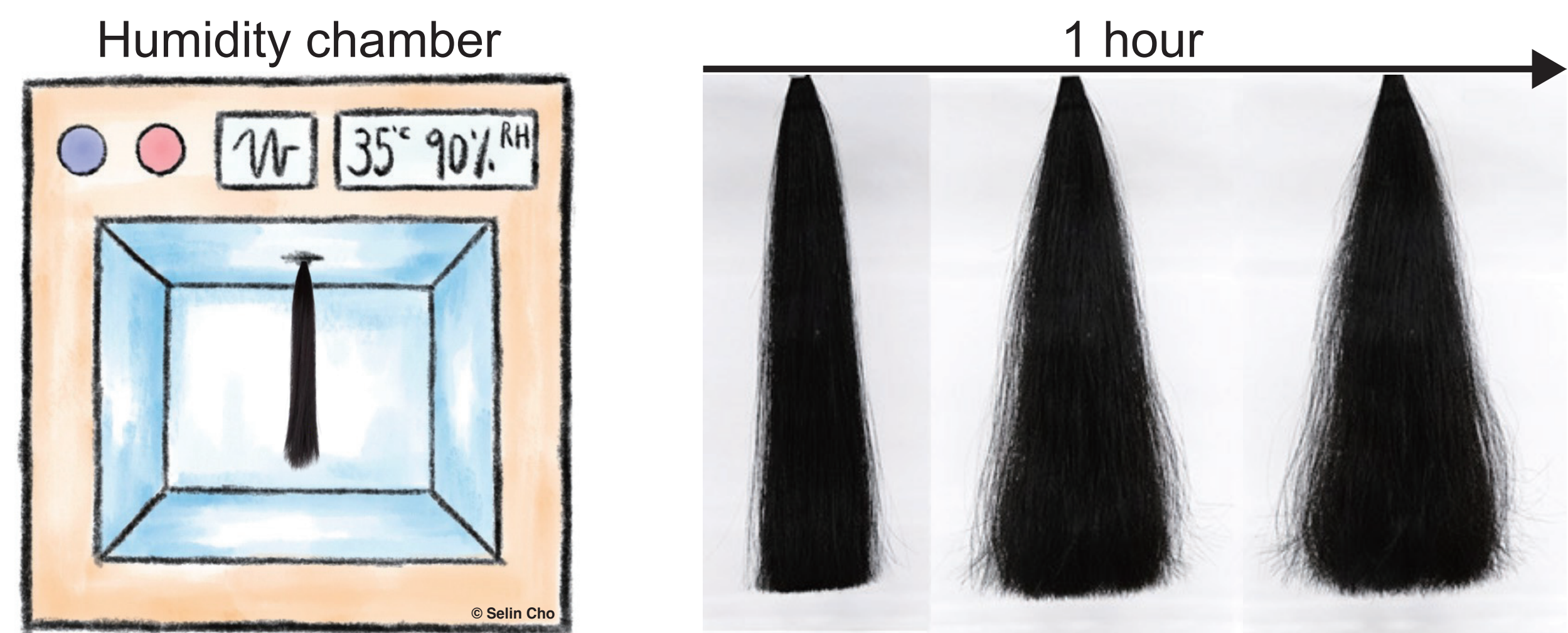
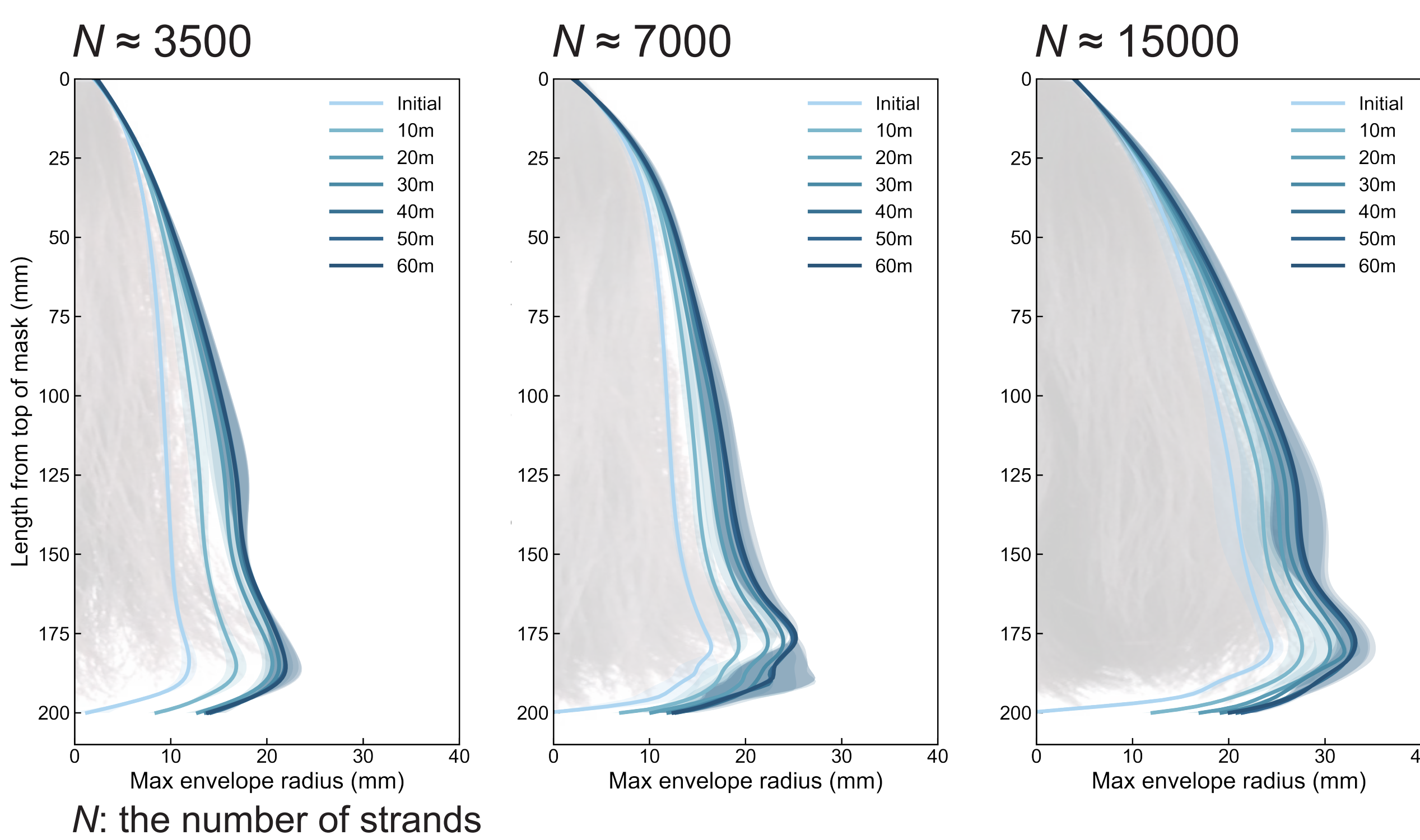
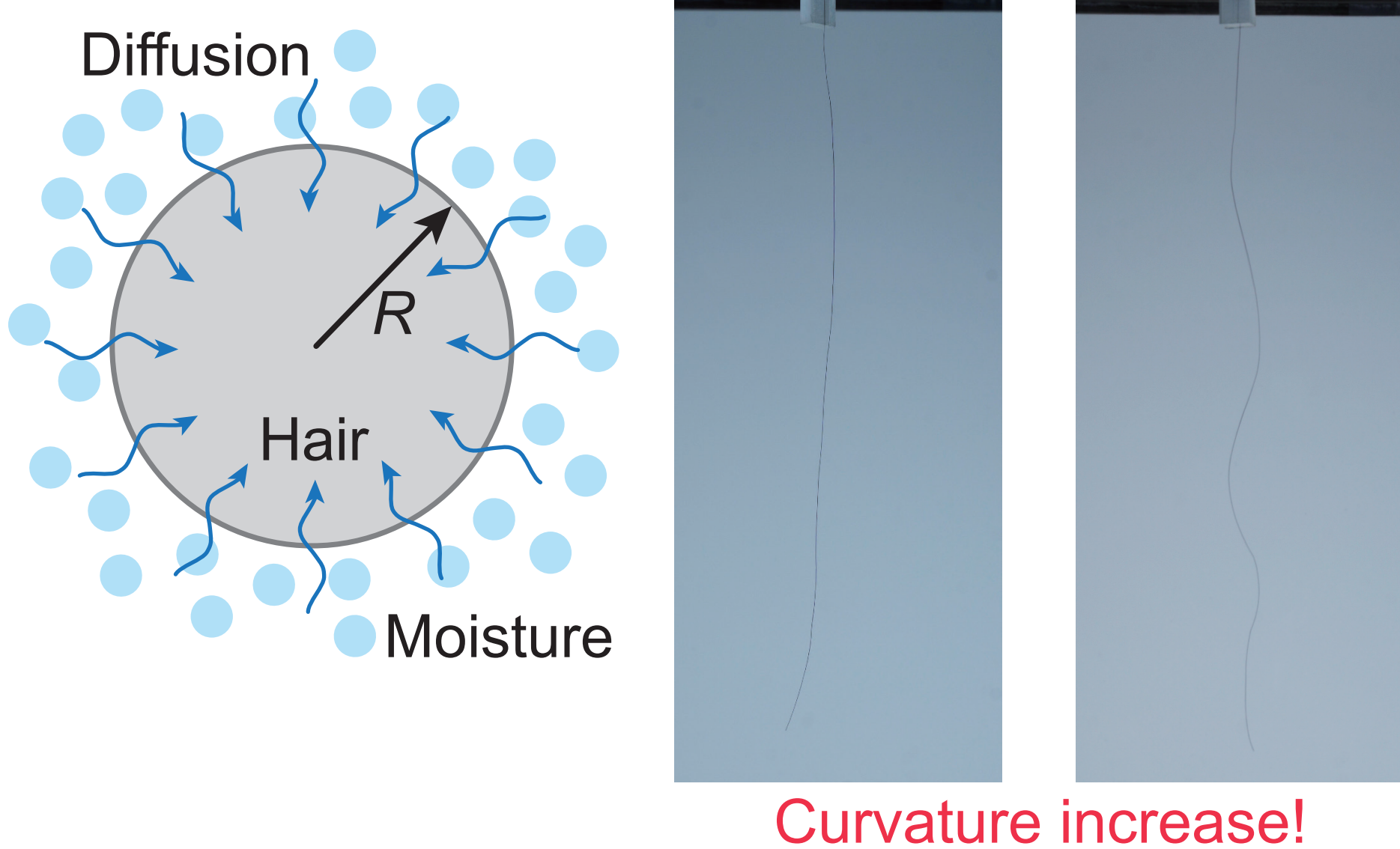


Image processing



Objectives

- We aim to elucidate the mechanics of moisture-driven hair frizz
- Our work sheds light on the bridge between mechanical engineering and beauty science.



Predict the frizz. Design the style.

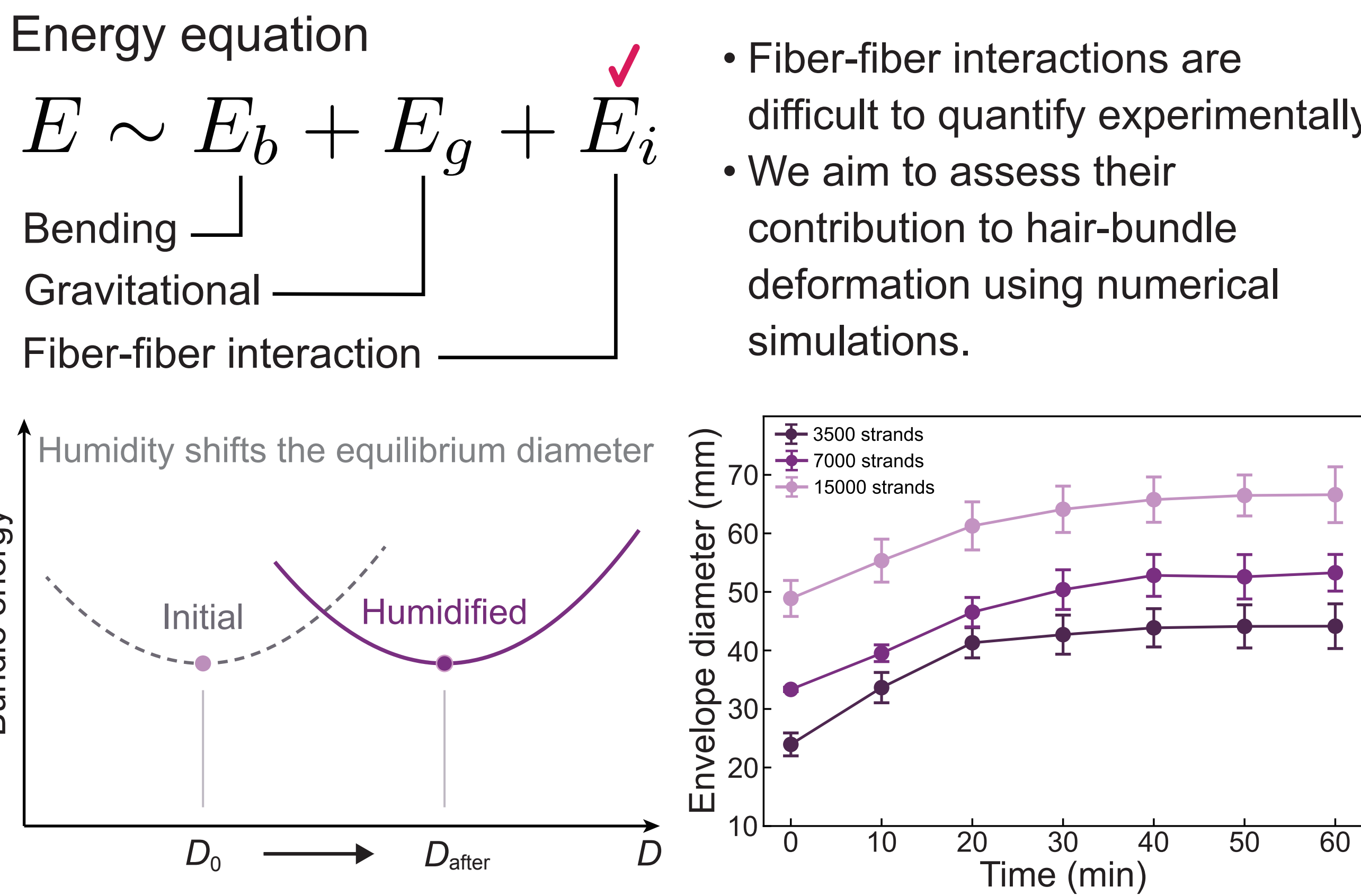
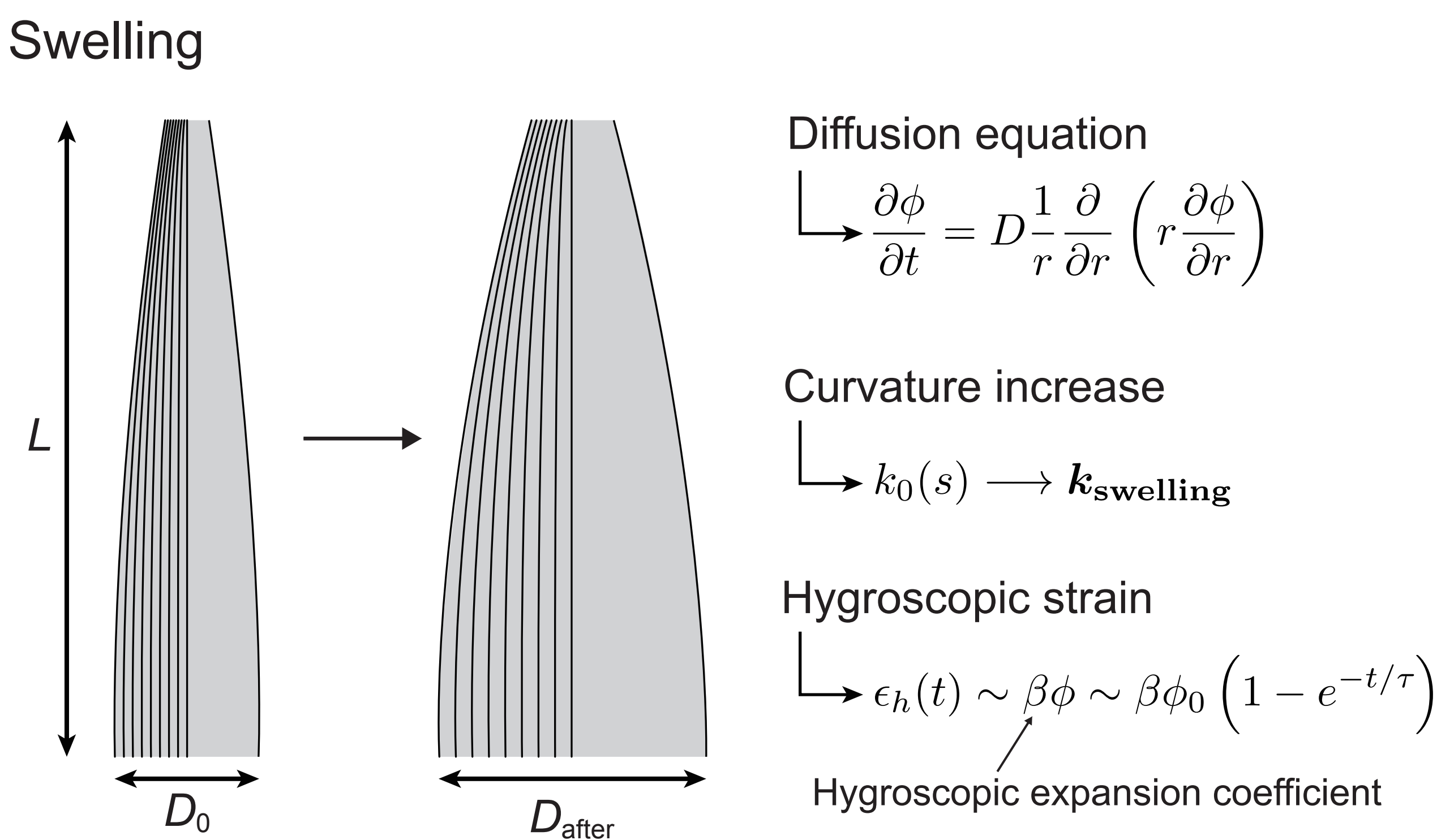
Acknowledgement

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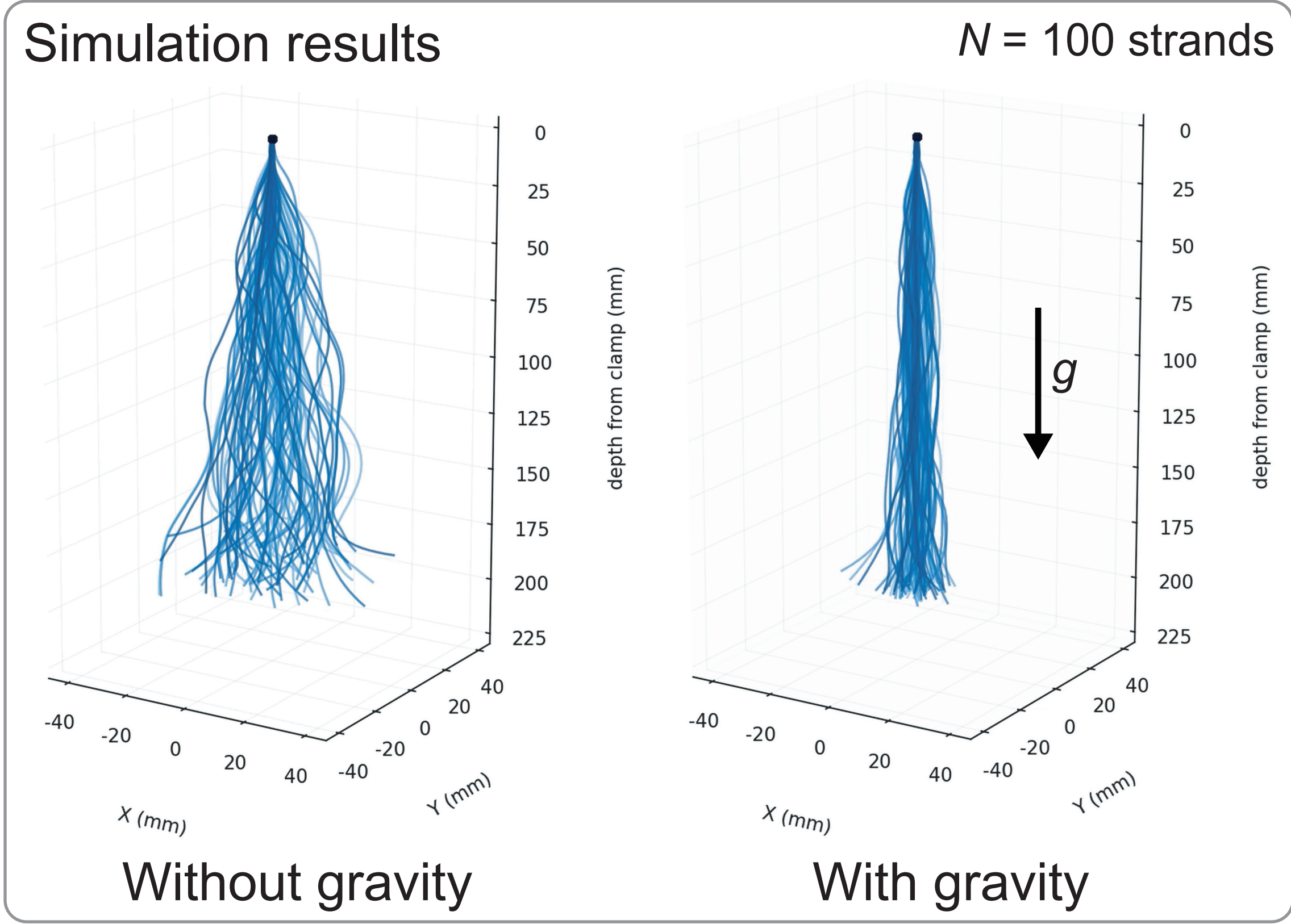
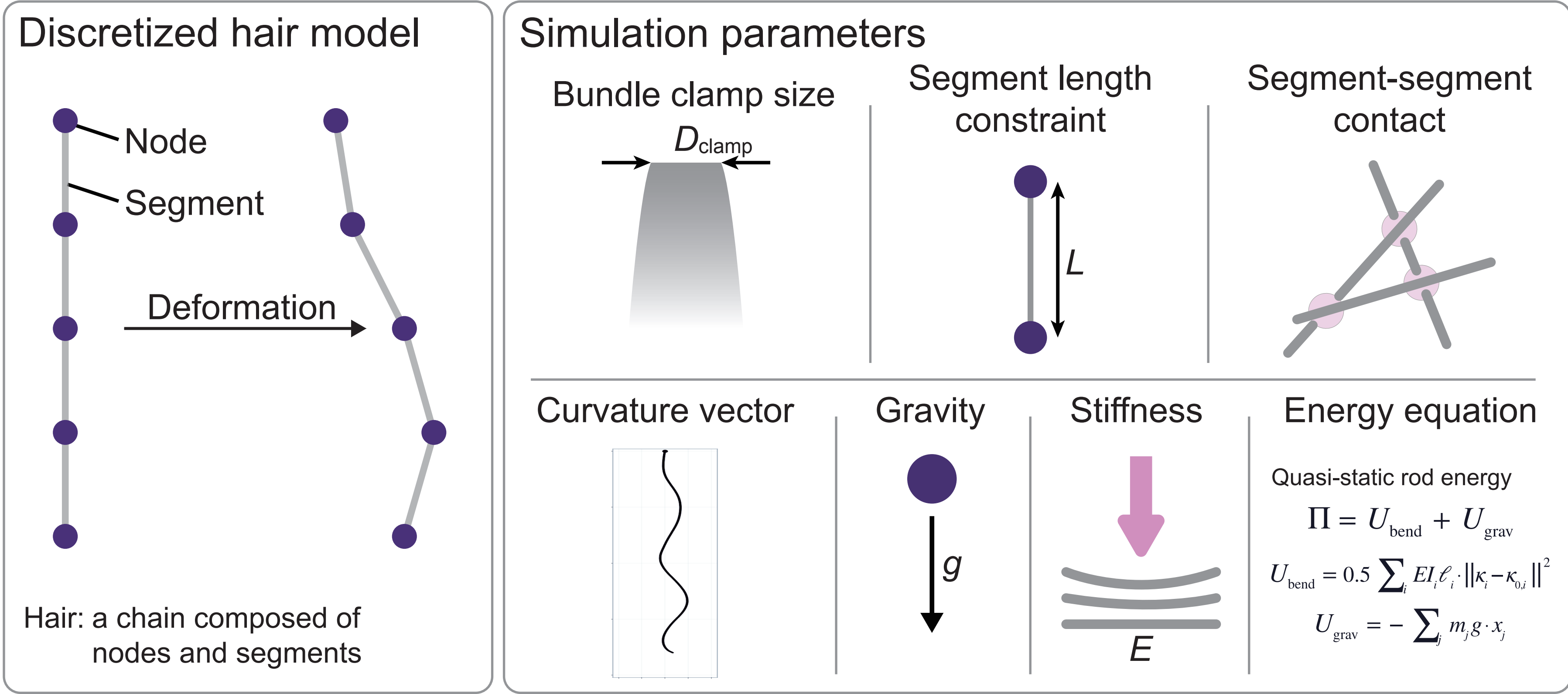
Supplementary video



Mechanical analysis



Numerical simulation



Conclusion

- Under humid conditions, hygroscopic swelling increases hair curvature, contributing to hair frizz.
- These findings connect mechanical engineering with beauty science and may inform future strategies for frizz control.